

Machine Learning in Business

MIS710 – A1

Assessment Task I – Case Study (Business Report) – Individual Predictive Analytics for Game Ratings at Play Quest Conquer



LEON WILSON
224142965
S224142965@deakin.edu.au



CONTENTS

Executive Summary	2
Business Understandings.....	2
Data Understanding, Preparation, Exploration, Visualization, and Insights.....	3
Machine Learning Model Development.....	5
Model Evaluation and Performance Metrics.....	5
Discussion of Pros and Cons of the Model.....	6
Business Solution and Recommendations	7
References	8

SECTION

Executive Summary

Play Quest Conquer (PQC) is a global online gaming platform based in Sydney, Australia, working to understand the key factors that influence game ratings. Gaining this understanding is essential for improving game development, acquisition, and marketing strategies, which are crucial for enhancing user satisfaction and boosting overall market performance.

To address this, a linear regression model was developed, focusing on key predictors such as game complexity, user interest, and popularity. The model successfully explained 43% of the variance in average ratings, indicating a moderate level of accuracy. The analysis highlighted that game complexity and user interest play significant roles in driving higher ratings, while game configuration factors like the number of players had a minimal impact.

To improve ratings, PQC should prioritize developing more complex games, leverage early user interest in marketing campaigns, increase game ownership through promotions and bundles, and expand game offerings for older audiences. Additionally, continuously refining the model with updated data will ensure it remains accurate and relevant. Implementing these strategies will help PQC enhance user satisfaction, expand its market reach, and strengthen its competitive position in the online gaming industry. Implementing these strategies will position PQC for long-term growth, increased market share, and sustained success.

Business understanding and Business Analysis Core Concept Model (BACCM) Elements:

The business problem Play Quest Conquer (PQC) faces is understanding the key factors that influence game ratings on its platform. This insight is crucial for making informed decisions in game development, acquisition, and promotion to enhance user satisfaction, optimize game offerings, and improve overall market performance. By accurately identifying and leveraging these factors, PQC can better align its strategies with user preferences, leading to higher engagement and a stronger competitive position in the global online gaming market.

- **Change:** PQC needs a deeper understanding of game rating influencers to make better decisions in development, acquisition, and promotion, enhancing user satisfaction and engagement.
- **Need:** Aligning game offerings with user preferences will optimize the gaming experience, leading to higher engagement and satisfaction. Data-driven insights will guide development and marketing strategies.
- **Solution:** Analyze the dataset to build a machine learning model that estimates game ratings, focusing on game configuration, engagement, and complexity, and providing strategic recommendations.
- **Stakeholders:** Market Research Manager, Game Development Team, Marketing Team, Executive Team

- **Value:** Aligning games with preferences increases user satisfaction, retention, and PQC's market position, boosting subscription renewals and game purchases.
- **Context:** PQC operates in a competitive global market; understanding diverse user preferences is crucial, and comprehensive data analysis is essential for actionable insights.

Data understanding, data preparation, exploration, visualization, and insights gained.

Data understanding: The dataset comprises 24,813 records, offering detailed insights into various game attributes available on the PQC platform.

Data Preparation:

- **Handling Missing Values:** Rows with missing values were removed to maintain dataset integrity, ensuring that the model was trained with complete data.
- **Outlier Detection and Treatment:** Outliers were identified using box plots and were replaced with median or mean values (for average playtime and average rating) to minimize their impact on the model.
- **Categorical Variable Encoding:** Categorical variables like Game_Type and Age_Category were converted into numerical formats using custom mappings based on category frequency to facilitate model training.

Data Exploration and Visualisation: Upon first glance, I was able to find that **Game_Name** had 7 missing variables. Since the dataset was large enough, removing the records with missing Game_Name was the obvious choice. Then was made sure that each feature was imported as correct datatypes. In order to understand each variable and Univariate analysis was conducted.

- Base Game dominate across all age categories, especially among younger players aged "5 to under 12" and "12 to under 18."
- Premium Games are more common in the "under 5" and "5 to under 12" groups, indicating a focus on younger children.
- Minimal presence of both game types in older age categories suggests an opportunity for PQC to expand offerings for older audiences.
- Game count: Base Game = 20,789; Premium Games = 4,017.
- The Released_Year data had 89 erroneous entries with "-99," which were dropped for cleaner analysis, as replacing them with the median (2015) was not logical.
- The age distribution is skewed toward younger audiences, particularly school-aged children.
- Most games require at least 2 players, though some allow for single-player gameplay.

- A few games require 3 or more players, catering to social or group play.
 - Majority of games support 2 to 5 players, with some exceptions allowing up to 100 players, likely representing large-scale or multiplayer experiences.
 - Overall, PQC's offerings are primarily geared toward younger audiences and smaller group play.
2. **Average play time:** The distribution of average playtime shows that most games on the platform have typical playtimes around 30, 45, 60, and 90 minutes, with 45 minutes being the most common. While there are games with very short or very long playtimes, most are designed for sessions lasting 30 to 60 minutes. Longer play sessions are less common, indicating the platform primarily caters to shorter, more manageable gaming experiences.
 3. **Relationship between the playing time and the average ratings:** The visualisation and a correlation coefficient of 0.22 indicate a weak positive relationship between average play time and average ratings. Although there is a slight tendency for games with longer play times to receive higher ratings, the connection is not particularly strong. This suggests that while play time may have some impact on ratings, other factors are likely more important in determining a game's rating on the PQC platform.
 4. **Relationship between level of game complexity and the average ratings:** The visualisation and a correlation coefficient of 0.44 indicate a moderate positive relationship between the level of game complexity and average ratings. This suggests that, generally, as the complexity of a game increases, its average rating tends to be higher. However, the relationship is not perfect, meaning that while complexity is an important factor in determining a game's rating, other factors likely contribute as well.
 5. **Game Configuration:** The correlation between game configuration variables and average ratings shows a weak influence. The minimum number of players required has a slightly negative correlation (-0.062) with ratings, suggesting that games needing fewer players might be rated slightly higher. The maximum number of players allowed has an almost negligible correlation (-0.0013), indicating that player capacity doesn't significantly affect ratings. These findings suggest that the number of players a game accommodates isn't a major factor in its success.
- Popularity:** Games with more owners tend to have higher ratings, as indicated by a moderate positive correlation (0.24). This suggests that broader ownership likely leads to better reviews, possibly due to positive word-of-mouth. Conversely, the number of traders shows a weaker correlation (0.092), implying that trading activity has less impact on ratings.

Interest: User interest is a significant factor in game ratings. Both "high interest" and general interest markers have a moderate positive correlation (0.43) with ratings. Games generating more user interest, whether marked as "high interest" or general interest, tend to receive higher ratings. Capturing and maintaining user interest is crucial for achieving high ratings on the platform.

Machine learning model development and evaluation.

A linear regression model was selected to predict Average Rating due to its simplicity and interpretability, helping PQC understand how game attributes affect ratings. A correlation matrix identified key features like Owner_Number, Trader_Number, HighInterest_Number, Min_Players, Max_Players, Interest_Number, mapped_game_type, Average_Complexity, Average_Play_Time, mapped_age, Complexity_Raters, and Rater_Number as predictors. Log transformations were applied to reduce outliers, and categorical variables were encoded numerically. Features with a correlation coefficient above 0.2 were prioritized for better predictive accuracy. The dataset was split into 80% training and 20% testing data to ensure the model generalizes well, avoiding overfitting and providing a reliable measure of accuracy.

Model Evaluation and Performance Metrics

Efforts to improve the model included scaling the values between 0 and 1 and adding polynomial features to enhance performance. However, these adjustments did not lead to any significant changes in the results.

Mean Absolute Error:	0.60
Root Mean Squared Error (RMSE):	0.8265504111254716
R-Squared (R^2):	0.42611029828068836

It achieved a Mean Absolute Error (MAE) of 0.60, indicating that, on average, the model's predictions were off by 0.60 rating points. The Root Mean Squared Error (RMSE) was 0.83, reflecting the average magnitude of prediction errors, with larger errors being more heavily penalized. The R-Squared (R^2) value was 0.43, meaning that approximately 43% of the variance in Average Ratings was explained by the model. The model shows moderate accuracy, there is room for improvement in capturing the factors that influence game ratings.

Pros and Cons of the Model

Pros

1. **Simplicity and Interpretability:** The linear regression model is easy to understand, making it clear how different game attributes affect ratings. This aids PQC in making informed decisions.
2. **Effective Feature Selection:** Using a correlation matrix to choose features ensures the model focuses on variables with a meaningful relationship to Average Rating, enhancing predictive accuracy.
3. **Handling Outliers:** Log transformations applied to key variables effectively reduced outliers, improving the model's robustness.
4. **Balanced Data Splitting:** The 80/20 train-test split helps prevent overfitting and provides a reliable measure of the model's performance.
5. **Focus on Popularity and Interest:** The model highlights the importance of user interest and game popularity, which show a moderate correlation with ratings, guiding PQC in prioritizing high-potential games.

Cons:

1. **Moderate Accuracy:** The model's R-Squared value of 0.43 indicates that it explains only 43% of the variance in Average Ratings, leaving much unexplained.
2. **Limited by Linear Relationships:** The model assumes linear relationships, which may not fully capture the complexities of game ratings, even with the addition of polynomial features.
3. **Sensitivity to Data Quality:** Data issues like typos in the Released_Year variable required dropping records, which could affect accuracy.
4. **Weak Influence of Some Features:** Game configuration variables like player numbers have a weak impact on ratings, suggesting other factors might be more significant.
5. **Room for Improvement:** More advanced models, such as random forests, could potentially improve predictive accuracy by capturing more complex relationships.

Business Solution and Recommendations

Business Solution: The linear regression model highlights how game complexity, user interest, and popularity influence ratings on Play Quest Conquer (PQC). By focusing on these key factors, PQC can make data-driven decisions in game development, acquisition, and marketing to enhance user satisfaction and platform success.

Recommendations for Implementation and Improvement:

1. **Focus on Game Complexity:** Prioritize developing or enhancing games with greater complexity, as these tend to receive higher ratings. Highlight the complexity in marketing to attract users seeking more engaging experiences.
2. **Leverage User Interest:** Track and analyze early interest in games. Prioritize promoting games that generate significant user interest, as they are more likely to receive high ratings. Encourage users to mark games as "high interest" to boost visibility.
3. **Increase Game Ownership:** Develop strategies to boost game ownership, such as offering bundles or promotions. Higher ownership is associated with better ratings, so engaging more players could improve overall ratings.
4. **Expand Offerings for Older Audiences:** Address the gap in games for older players by developing or acquiring titles tailored to this demographic, potentially attracting a broader audience.
5. **Refine the Model:** Regularly update the model with new data and explore advanced techniques to improve accuracy. This will ensure the model continues to provide relevant insights.
6. **Ensure Data Quality:** Implement processes to maintain clean, accurate data, which is crucial for reliable model predictions and decision-making.

Conclusion: By focusing on these key areas, PQC can enhance user satisfaction, expand its market reach, and strengthen its position in the online gaming industry.

References

- *What is Business Analysis Core Concept Model (BACCM)?* (n.d.). Wwww.knowledgehut.com.
<https://www.knowledgehut.com/blog/business-management/the-business-analysis-core-concept-model-baccm>
- W3 Schools. (2019). *Python Functions*. W3schools.com.
https://www.w3schools.com/python/python_functions.asp
- Google. (2024). *Gemini*. Gemini.google.com.
<https://gemini.google.com/app>

